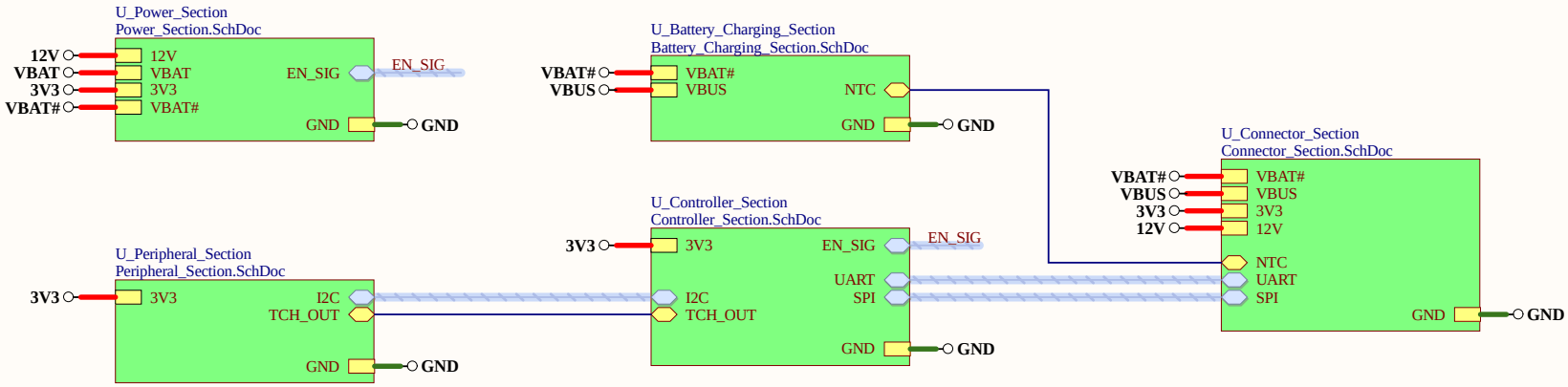
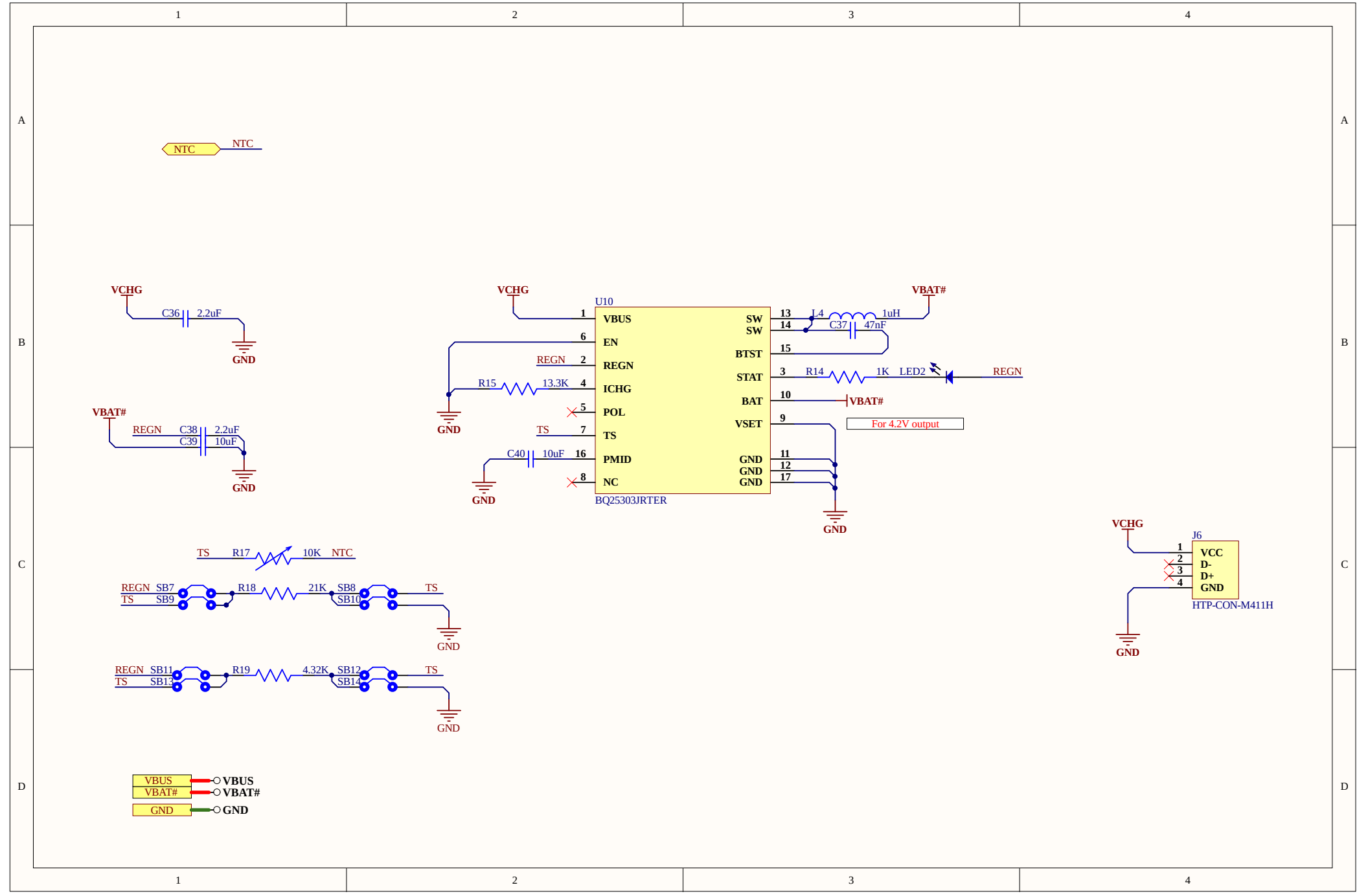
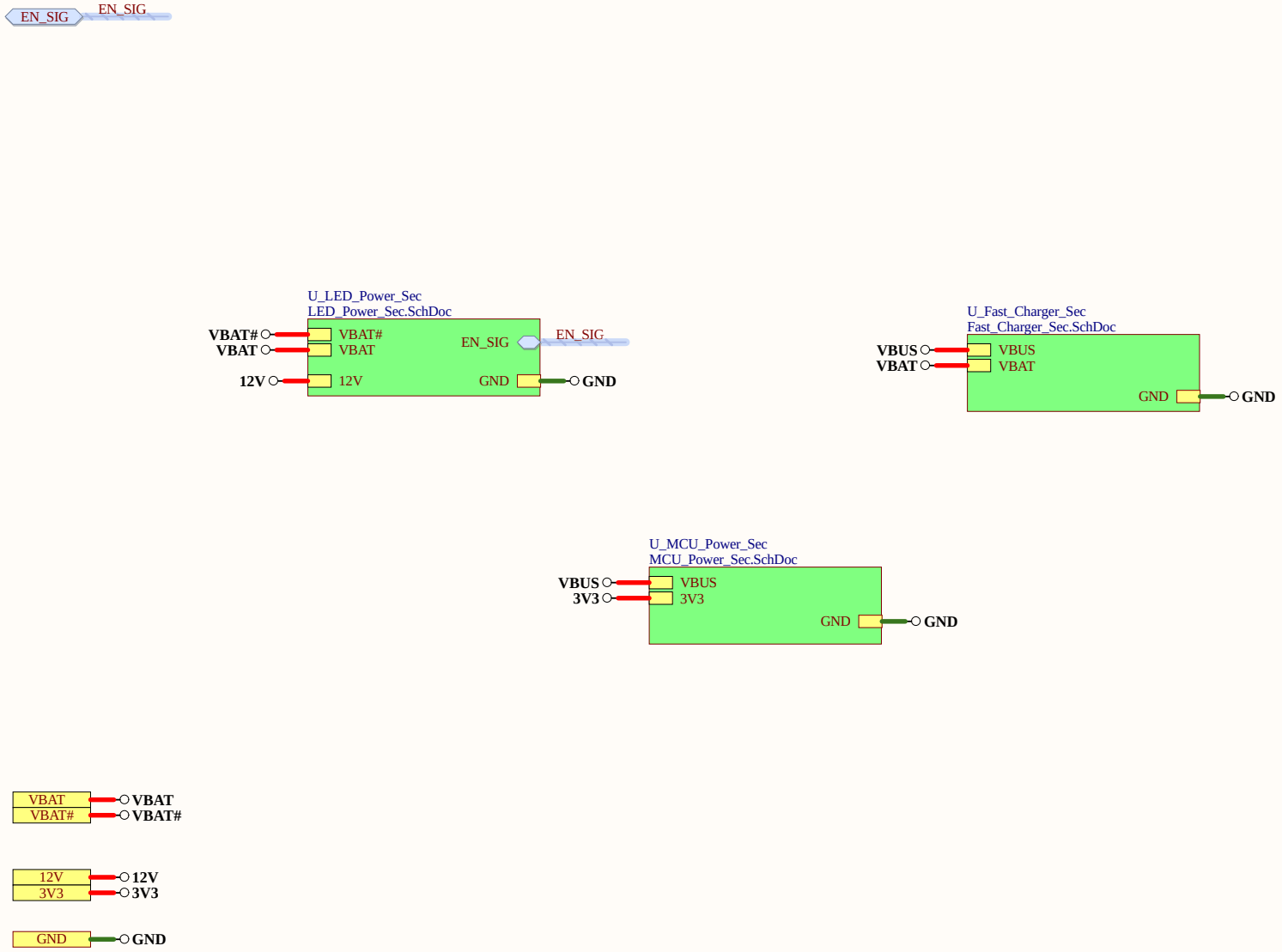
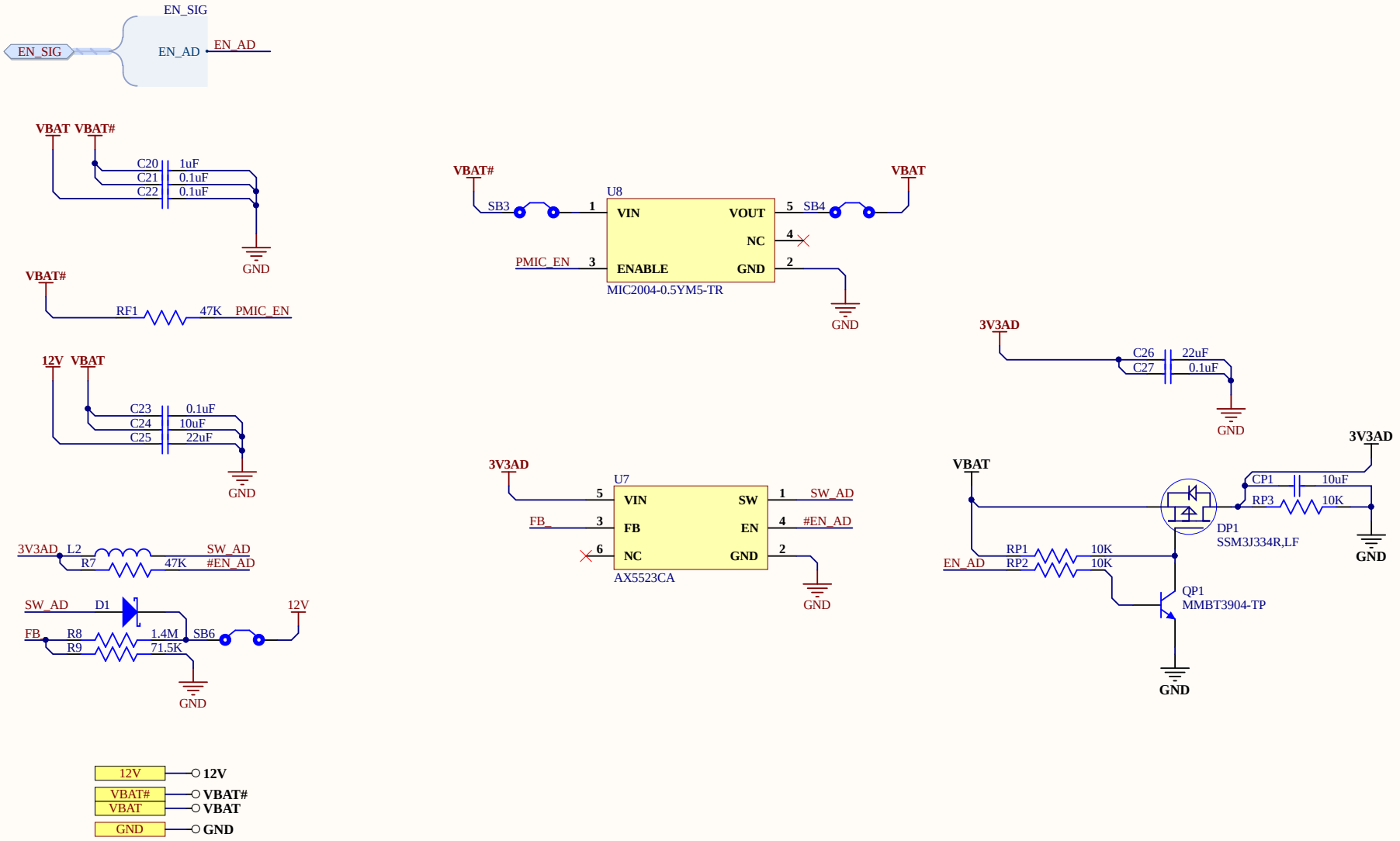
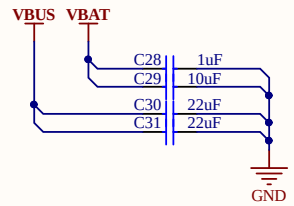
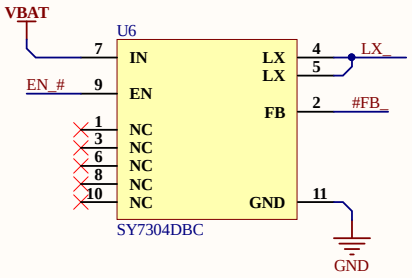

U_Power_Section
Power_Section.SchDoc
12V
VBAT
3V3
VBAT#
12V
VBAT
3V3
VBAT#
EN_SIG
EN_SIG
GND
GND
U_Battery_Charging_Section
Battery_Charging_Section.SchDoc
VBAT#
VBUS
VBAT#
VBUS
NTC
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VBAT#
VBUS
3V3
12V
VBAT#
VBUS
3V3
12V
NTC
UART
SPI
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
EN_SIG
EN_SIG
UART
SPI
I2C
TCH_OUT
I2C
TCH_OUT
GND
GND
U_Peripheral_Section
Peripheral_Section.SchDoc
3V3
3V3
I2C
TCH_OUT
GND
GND

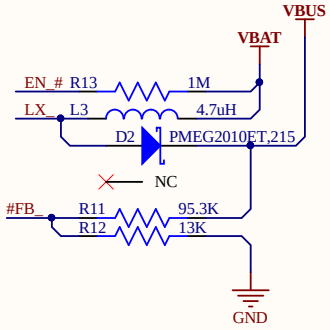

NTC
NTC
VCHG
C36
2.2uF
GND
VBAT#
REGN
C38
2.2uF
C39
10uF
GND
VCHG
U10
1
VBUS
6
EN
REGN
2
REGN
R15
13.3K
4
ICHG
5
POL
TS
7
TS
C40
10uF
16
PMID
8
NC
GND
BQ25303JRTER
13
SW
14
SW
15
BTST
3
STAT
10
BAT
9
VSET
11
GND
12
GND
17
GND
VBAT#
L4
1uH
C37
47nF
R14
1K
LED2
REGN
VBAT#
For 4.2V output
GND
TS
R17
10K
NTC
REGN
SB7
TS
SB9
R18
21K
SB8
SB10
TS
GND
REGN
SB11
TS
SB13
R19
4.32K
SB12
SB14
TS
GND
VCHG
J6
1
2
3
4
VCC
D-
D+
GND
HTP-CON-M411H
GND
VBUS
VBAT#
GND
VBUS
VBAT#
GND

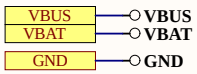

EN_SIG
EN_SIG
U_LED_Power_Sec
LED_Power_Sec.SchDoc
VBAT#
VBAT
12V
VBAT#
VBAT
12V
EN_SIG
GND
EN_SIG
GND
U_Fast_Charger_Sec
Fast_Charger_Sec.SchDoc
VBUS
VBAT
VBUS
VBAT
GND
GND
U_MCU_Power_Sec
MCU_Power_Sec.SchDoc
VBUS
3V3
VBUS
3V3
GND
GND
VBAT
VBAT#
VBAT
VBAT#
12V
3V3
12V
3V3
GND
GND

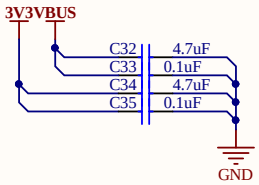

EN_SIG
EN_SIG
EN_AD
EN_AD
VBAT VBAT#
C20 1uF
C21 0.1uF
C22 0.1uF
GND
VBAT#
RF1 47K PMIC_EN
12V VBAT
C23 0.1uF
C24 10uF
C25 22uF
GND
3V3AD L2 SW_AD
R7 47K #EN_AD
SW_AD D1 12V
FB R8 1.4M SB6
R9 71.5K
GND
12V
VBAT#
VBAT
GND
12V
VBAT#
VBAT
GND
VBAT#
SB3
U8
VIN
VOUT
SB4
VBAT
NC
PMIC_EN
ENABLE
GND
MIC2004-0.5YM5-TR
GND
3V3AD
U7
VIN
SW
SW_AD
FB_
FB
EN
#EN_AD
NC
GND
AX5523CA
GND
3V3AD
C26 22uF
C27 0.1uF
GND
VBAT
3V3AD
CP1 10uF
RP3 10K
DP1
SSM3J334R,LF
GND
RP1 10K
EN_AD
RP2 10K
QP1
MMBT3904-TP
GND

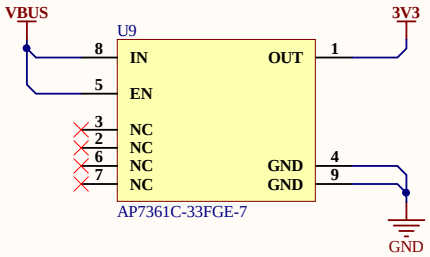

VBUS
VBAT
C28
1uF
C29
10uF
C30
22uF
C31
22uF
GND

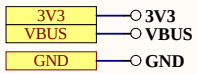

VBAT
U6
7
IN
EN_#
9
EN
1
NC
3
NC
6
NC
8
NC
10
NC
LX
4
LX
5
FB
2
#FB_
GND
11
SY7304DBC
LX_
GND

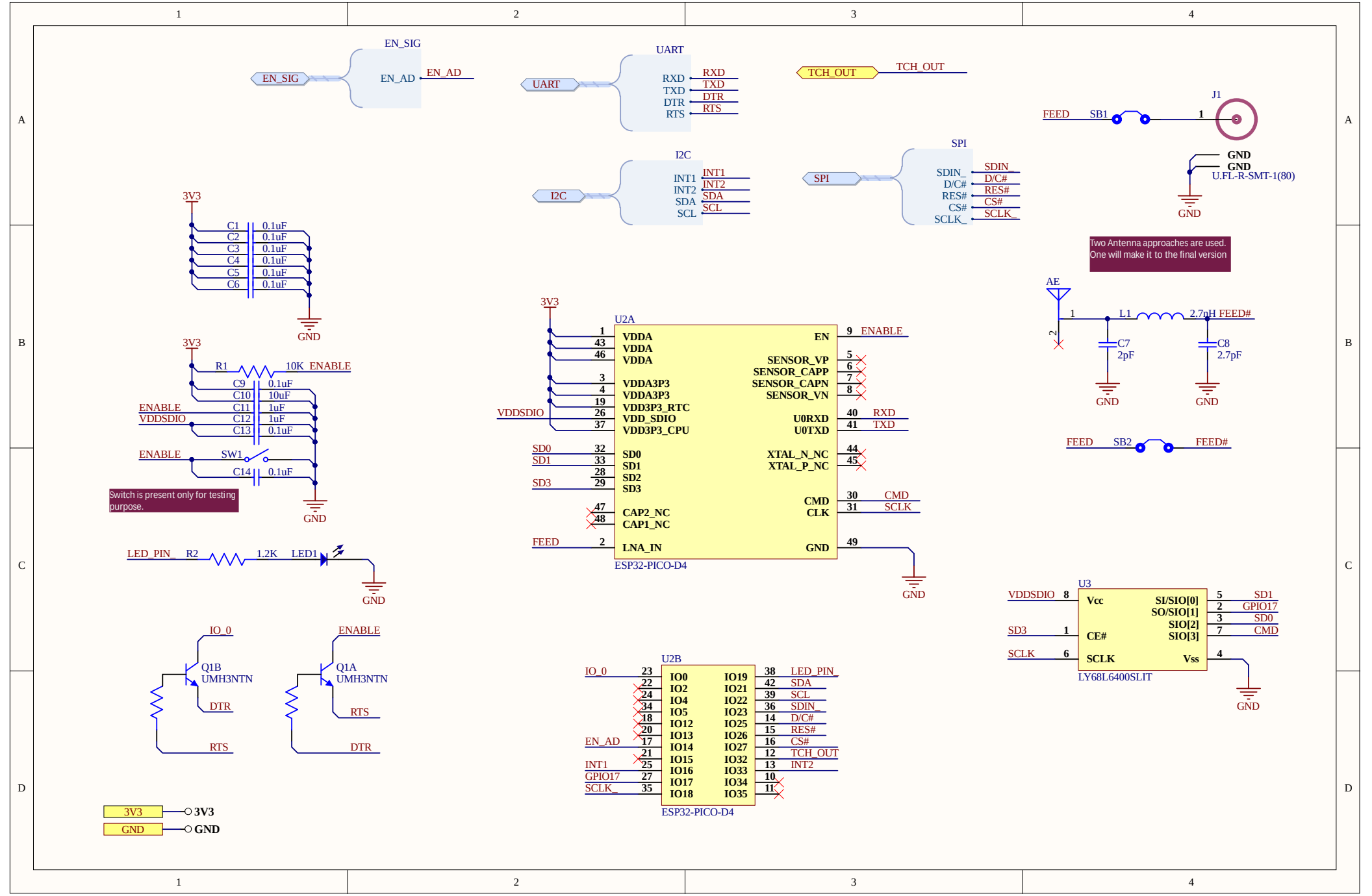

VBUS
VBAT
EN_#
R13
1M
LX_
L3
4.7uH
D2
PMEG2010ET,215
NC
#FB_
R11
95.3K
R12
13K
GND

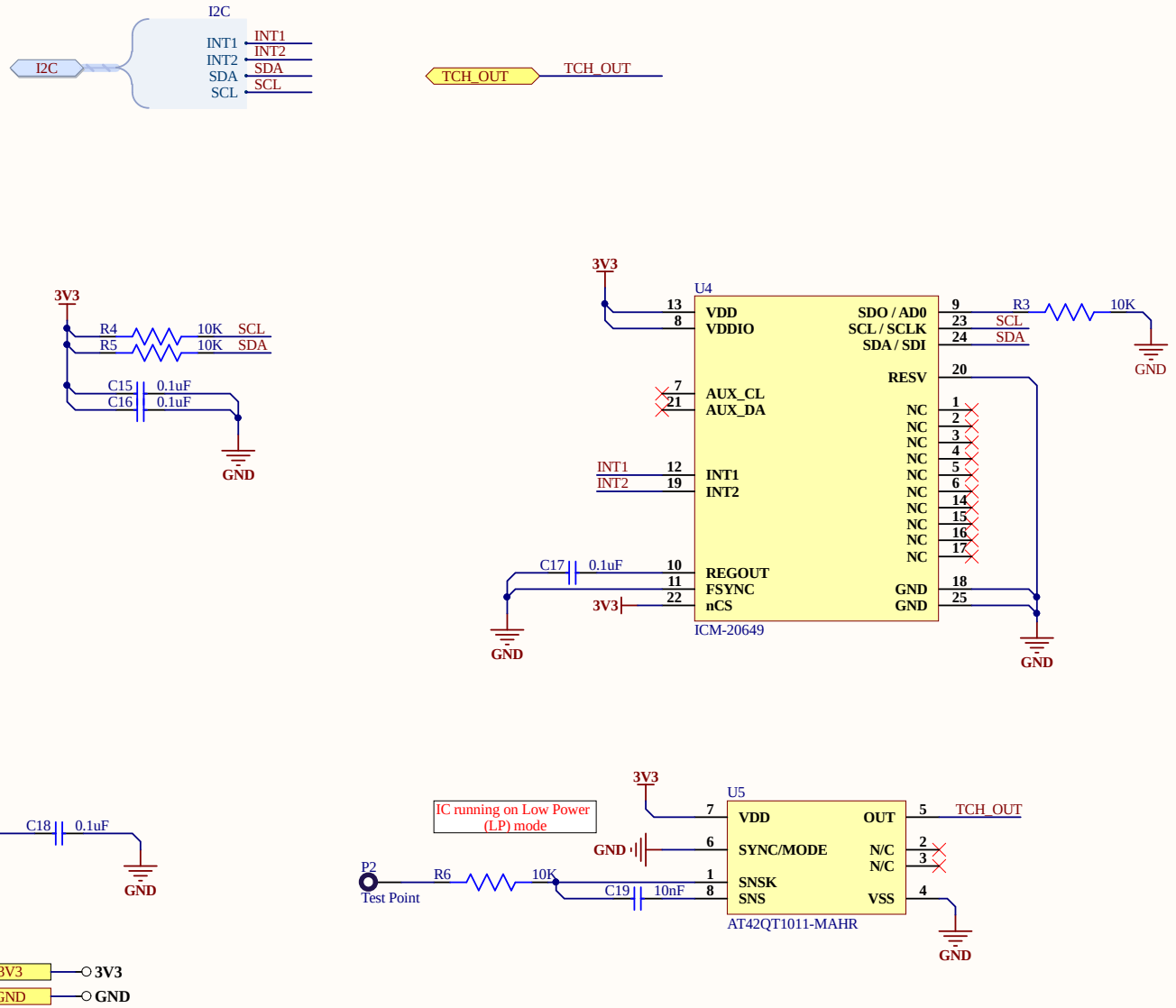

VBUS
VBAT
GND
VBUS
VBAT
GND

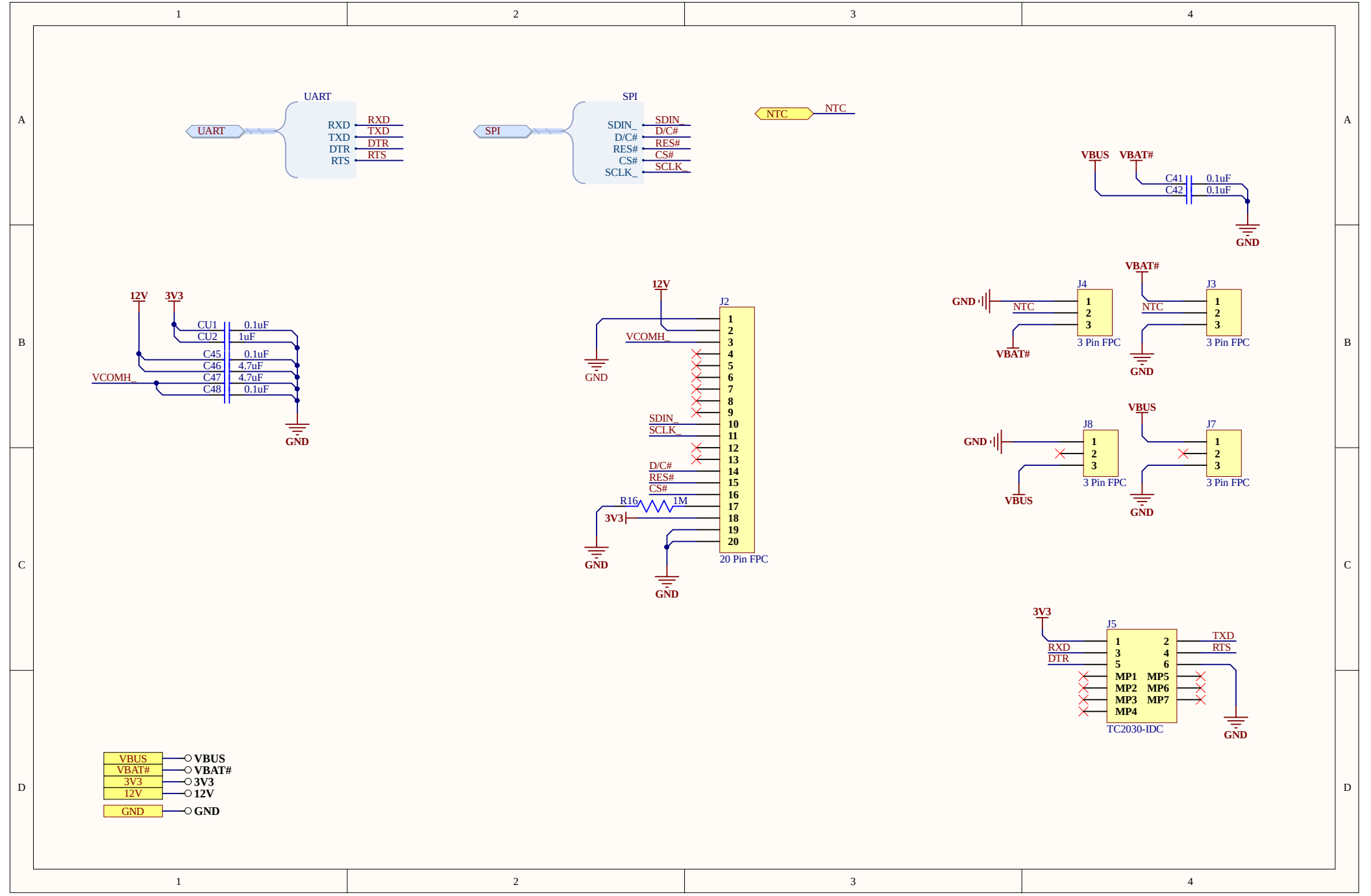

3V3
VBUS
C32
4.7uF
C33
0.1uF
C34
4.7uF
C35
0.1uF
GND


VBUS
3V3
U9
8
IN
OUT
1
5
EN
3
NC
2
NC
6
NC
GND
4
7
NC
GND
9
AP7361C-33FGE-7
GND


3V3
VBUS
GND
3V3
VBUS
GND


EN_SIG
EN_SIG
EN_AD
EN_AD
UART
UART
RXD
TXD
DTR
RTS
RXD
TXD
DTR
RTS
TCH_OUT
TCH_OUT
I2C
I2C
INT1
INT2
SDA
SCL
INT1
INT2
SDA
SCL
SPI
SPI
SDIN_
D/C#
RES#
CS#
SCLK_
SDIN_
D/C#
RES#
CS#
SCLK_
J1
FEED
SB1
1
GND
GND
U.FL-R-SMT-1(80)
GND
3V3
C1 0.1uF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
GND
Two Antenna approaches are used.
One will make it to the final version
AE
1
2
L1
2.7nH FEED#
C7
2pF
C8
2.7pF
GND
GND
3V3
R1
10K ENABLE
C9 0.1uF
C10 10uF
ENABLE
C11 1uF
VDDSDIO
C12 1uF
C13 0.1uF
ENABLE
SW1
C14 0.1uF
Switch is present only for testing purpose.
GND
FEED
SB2
FEED#
3V3
U2A
VDDA
VDDA
VDDA
VDDA3P3
VDDA3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
SD0
SD1
SD2
SD3
CAP2_NC
CAP1_NC
LNA_IN
EN
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
U0RXD
U0TXD
XTAL_N_NC
XTAL_P_NC
CMD
CLK
GND
ENABLE
RXD
TXD
CMD
SCLK
VDDSDIO
SD0
SD1
SD3
FEED
ESP32-PICO-D4
GND
LED_PIN_
R2
1.2K
LED1
GND
U3
VDDSDIO
Vcc
SI/SIO[0]
SO/SIO[1]
SIO[2]
SIO[3]
SD3
CE#
SCLK
SCLK
Vss
SD1
GPIO17
SD0
CMD
GND
LY68L6400SLIT
IO_0
Q1B
UMH3NTN
DTR
RTS
ENABLE
Q1A
UMH3NTN
RTS
DTR
U2B
IO_0
EN_AD
INT1
GPIO17
SCLK_
IO0
IO2
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
LED_PIN_
SDA
SCL
SDIN_
D/C#
RES#
CS#
TCH_OUT
INT2
ESP32-PICO-D4
3V3
3V3
GND
GND


I2C
INT1
INT2
SDA
SCL
TCH_OUT
3V3
R4
R5
10K
C15
C16
0.1uF
GND
U4
VDD
VDDIO
AUX_CL
AUX_DA
INT1
INT2
REGOUT
FSYNC
nCS
SDO / AD0
SCL / SCLK
SDA / SDI
RESV
NC
R3
C17
ICM-20649
C18
IC running on Low Power (LP) mode
U5
VDD
SYNC/MODE
SNSK
SNS
OUT
N/C
VSS
P2
Test Point
R6
C19
10nF
AT42QT1011-MAHR


1
2
3
4
A
B
C
D
UART
UART
RXD
TXD
DTR
RTS
RXD
TXD
DTR
RTS
SPI
SPI
SDIN_
D/C#
RES#
CS#
SCLK_
SDIN_
D/C#
RES#
CS#
SCLK_
NTC
NTC
VBUS
VBAT#
C41
0.1uF
C42
0.1uF
GND
12V
3V3
CU1
0.1uF
CU2
1uF
C45
0.1uF
C46
4.7uF
C47
4.7uF
C48
0.1uF
VCOMH_
GND
12V
J2
VCOMH_
GND
SDIN_
SCLK_
D/C#
RES#
CS#
R16
1M
3V3
GND
20 Pin FPC
GND
NTC
J4
3 Pin FPC
VBAT#
VBAT#
NTC
J3
3 Pin FPC
GND
GND
J8
3 Pin FPC
VBUS
VBUS
J7
3 Pin FPC
GND
3V3
J5
RXD
DTR
TXD
RTS
MP1
MP2
MP3
MP4
MP5
MP6
MP7
TC2030-IDC
GND
VBUS
VBAT#
3V3
12V
GND
VBUS
VBAT#
3V3
12V
GND